3.4.3 Rational Roots

If all the coefficients of a quadratic equation are integers, then Δ is an integer, and when it is positive, we have, $\sqrt{\Delta}$ is rational if, and only if, Δ is a perfect square. In other words, the equation $ax^2+bx+c=0$ with integer coefficients has rational roots, if, and only if, Δ is a perfect square.

What we discussed so far on polynomial equations of rational coefficients holds for polynomial equations with integer coefficients as well. In fact, multiplying the polynomial equation with rational coefficients, by a common multiple of the denominators of the coefficients, we get a polynomial equation of integer coefficients having the same roots. Of course, we have to handle this situation carefully. For instance, there is a monic polynomial equation of degree 1 with rational coefficients having $\frac{1}{2}$ as a root, whereas there is no monic polynomial equation of any degree with integer coefficients having $\frac{1}{2}$ as a root.

Example 3.11

Show that the equation $2x^2-6x+7=0$ cannot be satisfied by any real values of x.

Solution

$\Delta=b^2-4ac=-20<0$. The roots are imaginary numbers. ■

Example 3.12

If $x^2+2(k+2)x+9k=0$ has equal roots, find k.

Solution

Here $\Delta=b^2-4ac=0$ for equal roots. This implies $4(k+2)^2=4(9)k$. This implies $k=4$ or 1. ■

Example 3.13

Show that, if p,q,r are rational, the roots of the equation $x^2-2px+p^2-q^2+2qr-r^2=0$ are rational.

Solution

The roots are rational if $\Delta=b^2-4ac=(-2p)^2-4\left(p^2-q^2+2qr-r^2\right)$.

But this expression reduces to $4\left(q^2-2qr+r^2\right)$ or $4(q-r)^2$ which is a perfect square. Hence the roots are rational. ■

3.5 Applications of Polynomial Equation in Geometry

Certain geometrical properties are proved using polynomial equations. We discuss a few geometric properties here.

Example 3.14

Prove that a line cannot intersect a circle at more than two points.

Solution

By choosing the coordinate axes suitably, we take the equation of the circle as $x^2+y^2=r^2$ and the equation of the straight line as $y=mx+c$. We know that the points of intersections of the circle and the straight line are the points which satisfy the simultaneous equations

$$x^2+y^2=r^2 \qquad \ldots (1)$$

12th_Maths_EM_Vol1_CH 3_Theory_of_Eqn.indd 111 16-12-2022 12:38:44